

Chapter 9a Practice Test

Determine whether the sequence converges or diverges. If the sequence converges, find its limit.

1. $\left\{ \frac{n}{2n+1} \right\}$

2. $\left\{ \frac{n-4}{n^2+2} \right\}$

3. $\left\{ \frac{n^2+1}{3n(n+2)} \right\}$

4. $\left\{ \frac{6-n^{3/2}}{(\sqrt{n}+1)^2} \right\}$

5. $\left\{ (-1)^n \left(\frac{n^2-3}{2n^3-4} \right) \right\}$

6. $\left\{ (-1)^n \left(\frac{n^2-3}{4n^2-8} \right) \right\}$

Determine whether the given series converges or diverges. If the series converges and a sum can be found, find it. In the case of Alternating Series, classify the convergence as conditional or absolute.

7. $\sum_{n=1}^{\infty} \frac{4n^2-8}{n^2-5}$

8. $\sum_{k=1}^{\infty} 6 \left(\frac{2}{3} \right)^k$

9. $\sum_{k=1}^{\infty} \frac{1}{(3k+1)^2}$

10. $\sum_{k=0}^{\infty} \frac{1}{7^k}$

11. $\sum_{k=2}^{\infty} \frac{1}{k(\ln k)^2}$

12. $\sum_{k=2}^{\infty} 4 \left(\frac{2}{3} \right)^k$

13. $\sum_{k=1}^{\infty} 8 \left(-\frac{1}{7} \right)^{k+1}$

14. $\sum_{k=0}^{\infty} 4^k$

15. $\sum_{k=1}^{\infty} \frac{3k^2-6k+2}{4k^2+8}$

$$16. \sum_{n=1}^{\infty} \frac{(-1)^n 3n}{n^3 + 2}$$

$$17. \sum_{n=2}^{\infty} \frac{(-1)^{n+1}}{n!}$$

$$18. \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt[5]{n^3}}$$

$$19. \sum_{n=1}^{\infty} 4 \left(\frac{3^{n+1}}{5^{n-1}} \right)$$

$$20. \sum_{n=1}^{\infty} n^2 \sin \left(\frac{\pi n}{2} \right)$$

$$21. \sum_{k=0}^{\infty} k \left(\frac{2^k}{3^k} \right)$$

$$22. \sum_{k=1}^{\infty} \frac{k+1}{k}$$

$$23. \sum_{k=2}^{\infty} \frac{6}{k^2 - 1}$$

$$24. \sum_{n=1}^{\infty} \frac{(n+1)!}{3^n}$$

$$25. \sum_{n=1}^{\infty} \frac{n5^n}{(n+1)!}$$

$$26. \sum_{k=1}^{\infty} \frac{\arctan k}{k^2 + 1}$$

$$27. \sum_{n=1}^{\infty} \frac{(-1)^n n}{n^2 + 5}$$

28. Which of the following series converge?

$$\begin{array}{ll} \text{I.} & \frac{1}{e} + \frac{2}{e^2} + \frac{3}{e^3} + \frac{4}{e^4} + \dots \\ \text{II.} & 1 - \frac{1}{2} + \frac{1}{3} - \dots + \frac{(-1)^{n+1}}{n} + \dots \\ \text{III.} & 1 + \frac{1}{8} + \frac{1}{27} + \frac{1}{64} + \dots \end{array}$$

- a) None b) I and II c) I and III d) II and III e) I, II, III

29. Which of the following series converge(s)?

$$\begin{array}{ll} \text{I.} & 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} + \dots \\ \text{II.} & \frac{4}{1} + \frac{16}{2} + \frac{64}{6} + \frac{256}{24} + \dots \\ \text{III.} & 1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} - \dots + \frac{(-1)^{n+1}}{\sqrt{n}} + \dots \end{array}$$

- a) I only b) II only c) I and II d) II and III e) I, II and III

30. Which of the following sequences converge to 2?

a) $a_n = 2 + (-1)^n$ b) $a_n = \frac{2^n - 1}{n^2 + 3}$ c) $a_n = \frac{3 - 4n^2}{5 - 2n^2}$ d) $a_n = \frac{\sqrt{n^2 + 1}}{2n + 1}$ e) None of these

31. $\sum_{k=1}^{\infty} \frac{3^{k+1}}{4^k} =$

- a) 8 b) 9 c) 10 d) 12 e) 16

32. For which values of x does the series $\sum_{n=1}^{\infty} \frac{x^n(n-3)}{6^n}$ converge?

- a) $(-\infty, \infty)$ b) $(-3, 3)$ c) $(-6, 6)$ d) $(-3, 9)$ e) None of these

33. What are all values of p for which $\int_1^{\infty} \frac{1}{x^{2p}} dx$ converges?

- a) $p < -1$
- b) $p > 0$
- c) $p > \frac{1}{2}$
- d) $p > 1$
- e) There are no values of p for which this integral converges.

34. If the series $\sum_{n=1}^{\infty} a_n$ is conditionally convergent, determine which of the following series must diverge.

- a) $\sum_{n=1}^{\infty} a_n$
- b) $\sum_{n=1}^{\infty} |a_n|$
- c) $\sum_{n=1}^{\infty} (-1)^{2n} a_n$
- d) $\sum_{n=1}^{\infty} (-a_n)$
- e) None of these

35. Determine which of the following statements are true.

I. If $\lim_{n \rightarrow \infty} a_n = 0$, then $\sum_{n=1}^{\infty} a_n$ converges.

II. The Ratio Test cannot be used to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^3}$ converges.

III. If $a_n > 0$ and $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} (-1)^n a_n$ converges.

- a) None
- b) I and II
- c) I and III
- d) II and III
- e) I, II, III

36. Which of the following series is conditionally convergent?

- a) $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^{3/2}}$
- b) $\sum_{n=4}^{\infty} (-1)^n \frac{5n-3}{2n+6}$
- c) $\sum_{n=1}^{\infty} (-1)^n \frac{2n}{n^2+1}$
- d) $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{3^n}{n^4}$
- e) $\sum_{n=1}^{\infty} \frac{(-10)^n}{n!}$

37. The sum of the series $12 + 3 + \frac{3}{4} + \frac{3}{16} + \dots$ is

- a) 3
- b) 4
- c) 9
- d) 15
- e) 16

Answers

1. Since $\lim_{n \rightarrow \infty} \frac{n}{2n+1} = \lim_{n \rightarrow \infty} \frac{1}{2} = \frac{1}{2}$ by L'Hôpital, the sequence converges to $\frac{1}{2}$.
2. Since $\lim_{n \rightarrow \infty} \frac{n-4}{n^2+2} = \lim_{n \rightarrow \infty} \frac{1}{2n} = 0$ by L'Hôpital, the sequence converges to 0.
3. This can be simplified to $\frac{n^2+1}{3n^2+6n}$. Then, $\lim_{n \rightarrow \infty} \frac{n^2+1}{3n^2+6n} = \lim_{n \rightarrow \infty} \frac{2n}{6n+6} = \lim_{n \rightarrow \infty} \frac{2}{6} = \frac{1}{3}$ by L'Hôpital, the sequence converges to $\frac{1}{3}$.
4. Looking at the highest powers in the numerator and denominator gives us $n^{\frac{3}{2}}$ and $(\sqrt{n})^2 = n$, so $\lim_{n \rightarrow \infty} \frac{6-n^{\frac{3}{2}}}{(\sqrt{n}+1)^2} \approx \lim_{n \rightarrow \infty} \frac{-n^{\frac{3}{2}}}{n} = \lim_{n \rightarrow \infty} \frac{-\frac{3}{2}\sqrt{n}}{1} = -\infty$ by L'Hôpital and the sequence diverges.
5. Ignoring the alternating term at the beginning $\lim_{n \rightarrow \infty} \frac{n^2}{2n^3-4} = \lim_{n \rightarrow \infty} \frac{2n}{6n^2} = \lim_{n \rightarrow \infty} \frac{2}{12n} = 0$ by L'Hôpital. The alternating term will mean that the terms still approach 0, so the sequence converges to 0.
6. Ignoring the alternating term at the beginning $\lim_{n \rightarrow \infty} \frac{n^2-3}{4n^2-8} = \lim_{n \rightarrow \infty} \frac{2n}{8n} = \lim_{n \rightarrow \infty} \frac{2}{8} = \frac{1}{4}$ by L'Hôpital. The alternating term will mean that the terms will alternate between $\pm \frac{1}{4}$, so the sequence diverges.
7. The Nth term test for Divergence shows that $\lim_{n \rightarrow \infty} \frac{4n^2-8}{n^2-5} = \lim_{n \rightarrow \infty} \frac{8n}{2n} = \lim_{n \rightarrow \infty} \frac{8}{2} = 4$ by L'Hôpital. Since $4 \neq 0$, this series diverges.
8. This is a geometric series and because $\left|\frac{2}{3}\right| < 1$, this series converges. The sum of a geometric series is $S = \frac{a_1}{1-r}$ with the first term being $6\left(\frac{2}{3}\right)^1 = 4$. So $S = \frac{4}{1-\frac{2}{3}} = \frac{4}{\frac{1}{3}} = 12$.
9. This can be compared to $\sum_{k=1}^{\infty} \frac{1}{k^2}$, and since $p = 2 > 1$, this series converges.

10. This is a geometric series that could be rewritten as $\sum_{k=0}^{\infty} \left(\frac{1}{7}\right)^k$ and because $\left|\frac{1}{7}\right| < 1$, this series converges.

The sum of a geometric series is $S = \frac{a_1}{1-r}$ with the first term being $\left(\frac{1}{7}\right)^0 = 1$. So $S = \frac{1}{1-\frac{1}{7}} = \frac{1}{\frac{6}{7}} = \frac{7}{6}$.

11. This is an integral test, where $\int_2^{\infty} \frac{dk}{k(\ln k)^2}$. This requires a u-sub with $u = \ln k$, $du = \frac{dk}{k}$, so $\lim_{b \rightarrow \infty} \int_2^b \frac{du}{u^2}$.

So, $\lim_{b \rightarrow \infty} \int_2^b u^{-2} du = \lim_{b \rightarrow \infty} \left. \frac{-1}{\ln k} \right|_2^b = \lim_{b \rightarrow \infty} \left[\frac{-1}{\ln b} - \frac{-1}{\ln 2} \right]$. Since $\lim_{b \rightarrow \infty} \frac{-1}{\ln b} = 0$, the series converges (note that the series does not converge to the answer to the improper integral).

12. This is a geometric series and because $\left|\frac{2}{3}\right| < 1$, this series converges. The sum of a geometric series is

$S = \frac{a_1}{1-r}$ with the first term being $4\left(\frac{2}{3}\right)^2 = \frac{16}{9}$. So $S = \frac{\frac{16}{9}}{1-\frac{2}{3}} = \frac{\frac{16}{9}}{\frac{1}{3}} = \frac{16}{3}$.

13. This is a geometric series and because $\left|\frac{-1}{7}\right| < 1$, this series converges. The sum of a geometric series is

$S = \frac{a_1}{1-r}$ with the first term being $8\left(-\frac{1}{7}\right)^2 = \frac{8}{49}$. So $S = \frac{\frac{8}{49}}{1-\frac{-1}{7}} = \frac{\frac{8}{49}}{\frac{8}{7}} = \frac{1}{7}$.

14. This is a geometric series and because $|4| \geq 1$, this series diverges.

15. The Nth term test for Divergence shows that $\lim_{k \rightarrow \infty} \frac{3k^2 - 6k + 2}{4k^2 + 8} = \lim_{k \rightarrow \infty} \frac{6k - 6}{8k} = \lim_{k \rightarrow \infty} \frac{6}{8} = \frac{3}{4}$ by L'Hôpital.

Since $\frac{3}{4} \neq 0$, this series diverges.

16. This is the alternating series test, so two conditions must be true... 1) $\lim_{n \rightarrow \infty} \frac{3n}{n^3 + 2} = \lim_{n \rightarrow \infty} \frac{3}{3n^2} = 0$ by

L'Hôpital and 2) $\frac{3(n+1)}{(n+1)^3 + 2} < \frac{3n}{n^3 + 2}$, $3(n+1)(n^3 + 2) < 3n[(n+1)^3 + 2]$. The $(n+1)^3$ will cause the right side to be greater in comparison of the n , so since both conditions are true, this series converges.

To see if this absolutely converges, determine if $\sum_{n=1}^{\infty} \frac{3n}{n^3 + 2}$ converges. This can be compared to

$\sum_{n=1}^{\infty} \frac{n}{n^3} = \frac{1}{n^2}$, and since $p = 2 > 1$, this series converges and the alternating series converges absolutely.

17. This is the alternating series test, so two conditions must be true... 1) $\lim_{n \rightarrow \infty} \frac{1}{n!} = 0$ and

2) $\frac{1}{(n+1)!} < \frac{1}{n!}$, $n! < (n+1)!$, which is true. So, since both conditions are true, this series converges. To

see if this absolutely converges, determine if $\sum_{n=1}^{\infty} \frac{1}{n!}$ converges. We can use the ratio test, where

$\lim_{n \rightarrow \infty} \left| \frac{1}{(n+1)!} \cdot \frac{n!}{1} \right| = \lim_{n \rightarrow \infty} \left| \frac{n!}{(n+1)!} \right| = \lim_{n \rightarrow \infty} \left| \frac{1}{n+1} \right| = 0 < 1$, so this series converges and the alternating series **converges absolutely**.

18. This is the alternating series test, so two conditions must be true... 1) $\lim_{n \rightarrow \infty} \frac{1}{\sqrt[5]{n^3}} = 0$

and 2) $\frac{1}{(n+1)^{\frac{3}{5}}} < \frac{1}{n^{\frac{3}{5}}}$, $n^{\frac{3}{5}} < (n+1)^{\frac{3}{5}}$ which is true. So, since both conditions are true, this series

converges. To see if this absolutely converges, determine if $\sum_{n=1}^{\infty} \frac{1}{\sqrt[5]{n^3}}$ converges. Since $p = \frac{3}{5} \leq 1$, this series diverges and the alternating series **converges conditionally**.

19. This is a geometric series that could be rewritten as $\sum_{n=1}^{\infty} 4 \left(\frac{3^2 \cdot 3^{n-1}}{5^{n-1}} \right) = \sum_{n=1}^{\infty} 36 \left(\frac{3}{5} \right)^{n-1}$ and because $\left| \frac{3}{5} \right| < 1$,

this series **converges**. The sum of a geometric series is $S = \frac{a_1}{1-r}$ with the first term being $36 \left(\frac{3}{5} \right)^0 = 36$.

$$\text{So } S = \frac{36}{1 - \frac{3}{5}} = \frac{36}{\frac{2}{5}} = 90.$$

20. This is a disguised alternating series test because $\sin\left(\frac{\pi n}{2}\right)$ is going to alternate from ± 1 . So, two conditions must be true... 1) $\lim_{n \rightarrow \infty} n^2 = \infty$, which means that the alternating series test doesn't work.

However, by the Nth term test for divergence, since $\lim_{n \rightarrow \infty} n^2 \sin\left(\frac{\pi n}{2}\right) = DNE \neq 0$, this series **diverges**.

21. This is a mix of geometric and algebraic terms, so we should use the Ratio test. So

$$\lim_{k \rightarrow \infty} \left| \frac{(k+1) \left(\frac{2}{3} \right)^{k+1}}{k \left(\frac{2}{3} \right)^k} \right| = \lim_{k \rightarrow \infty} \left| \frac{2}{3} \cdot \frac{k+1}{k} \right| = \lim_{k \rightarrow \infty} \left| \frac{2}{3} \right| = \frac{2}{3} < 1 \text{ by L'Hôpital, so this } \text{converges}.$$

22. The Nth term test for Divergence shows that $\lim_{k \rightarrow \infty} \frac{k+1}{k} = \lim_{k \rightarrow \infty} \frac{1}{1} = 1$ by L'Hôpital. Since $1 \neq 0$, this series **diverges**.

23. This could use a comparison to $\sum_{k=2}^{\infty} \frac{1}{k^2}$ which is a p-series and since $p = 2 > 1$, this series converges.

24. This is a mix of geometric and factorial terms, so we should use the Ratio test. So

$$\lim_{n \rightarrow \infty} \left| \frac{(n+2)!}{3^{n+1}} \cdot \frac{3^n}{(n+1)!} \right| = \lim_{n \rightarrow \infty} \left| \frac{3^n (n+2)(n+1)!}{3 \cdot 3^n (n+1)!} \right| = \lim_{n \rightarrow \infty} \left| \frac{n+2}{3} \right| = \infty > 1, \text{ so this diverges.}$$

25. This is a mix of algebraic, geometric and factorial terms, so we should use the Ratio test. So

$$\lim_{n \rightarrow \infty} \left| \frac{(n+1)5^{n+1}}{(n+2)!} \cdot \frac{(n+1)!}{n5^n} \right| = \lim_{n \rightarrow \infty} \left| \frac{5 \cdot 5^n (n+1)(n+1)!}{5^n n(n+2)(n+1)!} \right| = \lim_{n \rightarrow \infty} \left| \frac{5(n+1)}{n(n+2)} \right| = \lim_{n \rightarrow \infty} \left| \frac{5}{2n+2} \right| = 0 < 1, \text{ so this converges.}$$

26. This is an integral test, where $\int_1^{\infty} \frac{\arctan k dk}{k^2 + 1}$. This requires a u-sub with $u = \arctan k$, $du = \frac{dk}{k^2 + 1}$, so

$$\lim_{b \rightarrow \infty} \int_1^b u du. \text{ So, } \lim_{b \rightarrow \infty} \int_1^b u du = \lim_{b \rightarrow \infty} \frac{1}{2} (\arctan k)^2 \Big|_1^b = \lim_{b \rightarrow \infty} \left[\frac{1}{2} (\arctan b)^2 - \frac{1}{2} (\arctan 1)^2 \right]. \text{ Since}$$

$$\lim_{b \rightarrow \infty} \frac{1}{2} (\arctan b)^2 = \frac{1}{2} \left(\frac{\pi}{2} \right)^2, \text{ the series converges (note that the series does not converge to this).}$$

27. This is the alternating series test, so two conditions must be true... 1) $\lim_{n \rightarrow \infty} \frac{n}{n^2 + 5} = \lim_{n \rightarrow \infty} \frac{1}{2n} = 0$

and 2) $\frac{n+1}{(n+1)^2 + 5} < \frac{n}{n^2 + 5}$, $(n^2 + 5)(n+1) < n[(n+1)^2 + 5]$ which is true because the squared term will cause the right side to be greater. So, since both conditions are true, this series converges. To see if this absolutely converges, determine if $\sum_{n=1}^{\infty} \frac{n}{n^2 + 5}$ converges. This can be compared to $\sum_{n=1}^{\infty} \frac{1}{n}$, which is the harmonic series, so this series diverges and the alternating series converges conditionally.

28. I. This can be written as $\sum_{n=1}^{\infty} \frac{n}{e^n}$. Since this is a mix of geometric and algebraic, we can use the ratio

$$\text{test. So, } \lim_{n \rightarrow \infty} \left| \frac{(n+1)}{e^{n+1}} \cdot \frac{e^n}{n} \right| = \lim_{n \rightarrow \infty} \left| \frac{e^n (n+1)}{e^1 \cdot e^n n} \right| = \lim_{n \rightarrow \infty} \left| \frac{n+1}{en} \right| = \lim_{n \rightarrow \infty} \left| \frac{1}{e} \right| = \frac{1}{e} < 1, \text{ meaning this converges.}$$

II. This can be written as $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$, which is the alternating harmonic series, which converges.

III. This can be written as $\sum_{n=1}^{\infty} \frac{1}{n^3}$. Seeing that $p = 3 > 1$, this converges, so E.

29. I. This can be written as $\sum_{n=1}^{\infty} \frac{1}{n}$, which is the harmonic series, which diverges.

II. This can be written as $\sum_{n=1}^{\infty} \frac{4^n}{n!}$. Since this is a mix of geometric and factorial, we can use the ratio

test. So, $\lim_{n \rightarrow \infty} \left| \frac{4^{n+1}}{(n+1)!} \cdot \frac{n!}{4^n} \right| = \lim_{n \rightarrow \infty} \left| \frac{4 \cdot 4^n n!}{4^n (n+1)n!} \right| = \lim_{n \rightarrow \infty} \left| \frac{4}{n+1} \right| = 0 < 1$, meaning this converges.

III. This can be written as $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}}$. This is an alternating series test, so two conditions must be

true... 1) $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} = 0$ and 2) $\frac{1}{\sqrt{n+1}} < \frac{1}{\sqrt{n}}$, $\sqrt{n} < \sqrt{n+1}$ which is true. So, since both conditions are true, this series converges. Therefore the answer is **D**.

30. Looking at a), $\lim_{n \rightarrow \infty} 2 + (-1)^n$ would oscillate between 1 and 3, meaning that the limit does not exist.

Looking at b) $\lim_{n \rightarrow \infty} \frac{2^n - 1}{n^2 + 3} = \frac{\infty}{\infty}$ meaning L'Hôpital applies. So, $\lim_{n \rightarrow \infty} \frac{2^n - 1}{n^2 + 3} = \lim_{n \rightarrow \infty} \frac{2^n (\ln 2)}{2n} = \frac{\infty}{\infty}$, so

L'Hôpital applies again. This gives us $\lim_{n \rightarrow \infty} \frac{2^n (\ln 2)}{2n} = \frac{2^n (\ln 2)^2}{2} = \infty$ so the limit does not exist. Looking

at c), $\lim_{n \rightarrow \infty} \frac{3 - 4n^2}{5 - 2n^2} = \frac{\infty}{\infty} = \lim_{n \rightarrow \infty} \frac{-8n}{-4n} = 2$ after a round of L'Hôpital, so **C**.

31. This is a geometric series that could be rewritten as $\sum_{k=1}^{\infty} \left(\frac{3^1 \cdot 3^k}{4^k} \right) = \sum_{k=1}^{\infty} 3 \left(\frac{3}{4} \right)^k$ and because $\left| \frac{3}{4} \right| < 1$, this

series converges and the sum of a geometric series is $S = \frac{a_1}{1-r}$ with the first term being $3 \left(\frac{3}{4} \right)^1 = \frac{9}{4}$. So

$$S = \frac{\frac{9}{4}}{1 - \frac{3}{4}} = \frac{\frac{9}{4}}{\frac{1}{4}} = 9 \text{ so } \mathbf{B}.$$

32. The series $\sum_{n=1}^{\infty} \frac{x^n (n-3)}{6^n}$ contains a mixture of geometric and algebraic, so we should use the Ratio Test.

So, $\lim_{n \rightarrow \infty} \left| \frac{x^{n+1} (n-2)}{6^{n+1}} \cdot \frac{6^n}{x^n (n-3)} \right| = \lim_{n \rightarrow \infty} \left| \frac{x \cdot x^n \cdot 6^n \cdot (n-2)}{6 \cdot x^n \cdot 6^n \cdot (n-3)} \right| = \lim_{n \rightarrow \infty} \left| \frac{x(n-2)}{6(n-3)} \right| = \frac{\infty}{\infty}$. This means we must apply

L'Hôpital (remembering that n is the constant) or $\lim_{n \rightarrow \infty} \left| \frac{x(n-2)}{6(n-3)} \right| = \lim_{n \rightarrow \infty} \left| \frac{x}{6} \right| = \left| \frac{x}{6} \right|$. According to the Ratio

Test, this will converge if the ratio is less than 1. $\left| \frac{x}{6} \right| < 1$, $-1 < \frac{x}{6} < 1$, $-6 < x < 6$ so **C**.

33. This integral is comparable to the series $\sum_{n=1}^{\infty} \left(\frac{1}{n^{2p}} \right)$ and this is a p – series which will converge when

$$2p > 1, p > \frac{1}{2}, \text{ so C.}$$

34. If $\sum_{n=1}^{\infty} a_n$ is conditionally convergent, that means $\sum_{n=1}^{\infty} a_n$ converges, but $\sum_{n=1}^{\infty} |a_n|$ diverges. So B.

35. I. This is not true, the harmonic series $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$ and $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

II. The ratio test would yield $\lim_{n \rightarrow \infty} \left| \frac{(n+1)^3}{n^3} \right| = \lim_{n \rightarrow \infty} \left| \frac{3(n+1)^2}{3n^2} \right| = \lim_{n \rightarrow \infty} \left| \frac{6(n+1)}{6n} \right| = \lim_{n \rightarrow \infty} \left| \frac{6}{6} \right| = 1 = 1$, which is inconclusive, so this is true.

III. If $\sum_{n=1}^{\infty} a_n$ converges, then adding the alternating term $(-1)^n$ will cause the sum to be smaller. So by the comparison test, $\sum_{n=1}^{\infty} (-1)^n a_n$ will converge, so true. Therefore, D.

36. A series is conditionally convergent if the alternating series converges, but the absolute value diverges.

So for a), $\lim_{n \rightarrow \infty} \frac{1}{n^{\frac{3}{2}}} = 0$ and 2) $\frac{1}{(n+1)^{\frac{3}{2}}} < \frac{1}{n^{\frac{3}{2}}}$, $n^{\frac{3}{2}} < (n+1)^{\frac{3}{2}}$ which is true. So, this series converges and

$\sum_{n=1}^{\infty} \frac{1}{n^{\frac{3}{2}}}$ converges since $p = \frac{3}{2} > 1$, this series converges absolutely. For b), The Nth term test for

divergence shows $\lim_{n \rightarrow \infty} \frac{5n-3}{2n+6} = \lim_{n \rightarrow \infty} \frac{5}{2} = \frac{5}{2} \neq 0$ by L'Hôpital, so this diverges. For c),

$\lim_{n \rightarrow \infty} \frac{2n}{n^2+1} = \lim_{n \rightarrow \infty} \frac{2}{2n} = 0$ by L'Hôpital, and 2) $\frac{2(n+1)}{(n+1)^2+1} < \frac{2n}{n^2+1}$, $2(n+1)(n^2+1) < 2n[(n+1)^2+1]$ which

is true. So, this series converges and $\sum_{n=1}^{\infty} \frac{2n}{n^2+1}$ can be compared to $\sum_{n=1}^{\infty} \frac{1}{n}$, which diverges, so C.

37. This is a geometric series with $r = \frac{1}{4}$, $a_1 = 12$. The sum of a geometric series is $S = \frac{a_1}{1-r}$, so

$$S = \frac{12}{1 - \frac{1}{4}} = \frac{12}{\frac{3}{4}} = 16 \text{ so E.}$$